APPENDIX E

Chapter 5
Aligned Minds, Efficient Machines:
Integrating Neuromorphic Computing for Personalized AI

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ABSTRACT

AI systems exhibit robust reasoning abilities yet face challenges in personalization and sustainability, as cognitive augmentation technologies often show rigidity and high energy use that limit effectiveness. This paper introduces a framework that integrates psychometric alignment, modular cognitive agents, and neuromorphic computing to deliver scalable, personalized, energy-efficient augmentation. The Synthetic Cognitive Augmentation Network Alignment Questionnaire (SCANAQ) captures users' cognitive inclinations (Tate, 2024a) and informs the Synthetic Cognitive Augmentation Network Using Experts (SCANUE), a modular system modeled on prefrontal-cortex subregions (Tate, 2024b). The Spiking Transformer Augmenting Cognition (STAC) pipeline targets neuromorphic edge deployment: STAC V1 demonstrates a hybrid fine-tuning approach, and STAC V2 provides an ANN-to-SNN conversion pathway that yields spiking-compatible networks (Tate, 2025a; Tate, 2025b). Future work includes empirical hardware validation and standardized neuromorphic benchmarking (Davies et al., 2021; Ostrau et al., 2022).

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INTRODUCTION

Consider an AI assistant that not only aids in critical medical diagnoses or financial decisions but also instinctively aligns its guidance with a user's individual decision-making style and emotional state. This vision remains largely unrealized despite increasing deployments of large-language models (LLMs) in high-stakes contexts such as healthcare, finance, and education. Although these models demonstrate impressive capabilities, two crucial, interconnected limitations significantly impede their potential. First, current LLMs typically lack deep personal alignment, operating instead with generic, one-size-fits-all response paradigms. Effective cognitive augmentation, however, necessitates a system that can dynamically adapt its interactions, reasoning strategies, and communication styles to match each user's unique and stable cognitive, affective, and experiential characteristics. Without this fine-grained personalization—which requires capturing nuanced individual profiles, including decision-making tendencies, affective regulation strategies, and real-time stress levels—user trust can diminish, engagement declines, and the system risks offering support that is misaligned or even detrimental to the user's goals. Fine-grained personalization therefore remains the decisive hurdle for real-world augmentation.

Second, energy efficiency presents a formidable and urgent obstacle. Training and deploying large-scale transformer-based models requires substantial computational resources, which has raised profound sustainability concerns. The energy demands and associated environmental impacts, as documented by Strubell et al. (2019), underscore the need to develop more sustainable computational paradigms. This intensive computational footprint contributes to a significant carbon footprint and restricts the deployment of sophisticated cognitive assistants on resource-limited edge devices, such as mobile phones or embedded systems. Such a barrier limits equitable, widespread access to powerful AI technologies, thereby creating a digital divide. To address this crisis, neuromorphic computing has emerged as a promising solution. Its brain-inspired, event-driven architectures compute only when new information is available, resulting in an inherently lower energy profile (Roy et al., 2019).

The intersection of these two constraints forms the central tension this research addresses. Any attempt to personalize AI through straightforward methods, such as independently running multiple large, fine-tuned models for different user profiles, would exponentially increase computational demands and exacerbate the already critical energy-consumption problem. Conversely, prioritizing energy efficiency by drastically reducing model complexity often sacrifices the depth and precision required for sophisticated and meaningful personalization. Thus, a truly viable cognitive augmentation framework must simultaneously optimize for both personalized alignment and computational sustainability. This paper proposes a structured, empirically evaluable solution by synthesizing three complementary lines of inquiry into a unified, integrated approach. Having established the core problem, the following sections will delve into the existing literature to frame the technical contributions of this work.

LITERATURE REVIEW

The Dual Challenges of Modern AI: Personal Alignment and Computational Sustainability

The advancement of AI faces two critical, interdependent challenges: achieving deeper personal alignment by enhancing meaningful human-AI interactions and ensuring computational sustainability by addressing the high-energy demands of advanced AI. The promise of cognitive augmentation—AI systems that not only provide information but actively enhance human thought processes—hinges entirely on resolving the tension between these goals. A system that is perfectly personalized but environmentally and economically unviable remains a theoretical curiosity; a system that is efficient but generically impersonal fails to deliver on the core promise of augmentation. This section, therefore, reviews the academic and engineering context of these two challenges to establish the theoretical and practical foundations upon which the SCANUE and STAC frameworks are built.

The Challenge of Static Alignment in Large Language Models

The dominant paradigm of large language models (LLMs) is built on the principle of generalism. Models based on architectures like the Transformer (Vaswani et al., 2017) are trained on vast, heterogeneous cross-sections of public data to perform a wide range of tasks for a generic, statistically average user. While this approach has led to remarkable capabilities in language understanding, generation, and reasoning, it has also created a significant "alignment gap." Current LLMs lack deep personal alignment, a limitation that manifests in several critical ways that impede their effective use as cognitive partners.

One primary manifestation is the erosion of trust. Research in human-computer interaction and psychology consistently shows that trust is predicated on predictability, reliability, and a sense of shared understanding. When an AI assistant's reasoning style, values, or communication patterns deviate sharply or unpredictably from the user's own cognitive model, it creates dissonance and alienates the user, undermining their confidence in the system (Tate, 2024b). This phenomenon is often described as model brittleness, where a system that performs well on average may fail unexpectedly when faced with an individual's unique cognitive style or cultural context. The one-size-fits-all approach necessarily smooths over the very individual nuances critical for high-stakes decision-making.

A second consequence is that the lack of personalization directly impedes the development of effective cognitive augmentation. True augmentation is not merely about providing correct answers or automating tasks; it is about scaffolding a user's own cognitive processes, enhancing their strengths, and supporting their weaknesses in a dynamic, context-aware manner (Miller & Cohen, 2001). This capability requires a system that can build and maintain a sophisticated model of the user, understanding their executive function strengths and weaknesses, their decision-making tendencies under stress, and their preferred methods of emotional regulation (Damasio, 1994).

A generic model cannot provide this nuanced support. It risks delivering advice that is not only misaligned but potentially detrimental to the user's goals—for instance, by encouraging a risk-averse user to take a gamble they are not psychologically equipped to handle, or by presenting information in a way that overwhelms a user with executive function challenges. The SCANUE framework directly addresses this fundamental challenge by proposing that deep alignment is not a post-hoc feature but a foundational architectural principle (Kennedy, Mohammad, & Wyandt, 2025). Consequently, this alignment framework operationalizes a user's rich, stable psychometric profile, mapping it directly onto a system of specialized cognitive agents designed to mirror the functional organization of the human prefrontal cortex.

The Sustainability Crisis and the Promise of Neuromorphic Computing

The computational demands of training and deploying state-of-the-art LLMs have created a sustainability crisis. The seminal work of Strubell et al. (2019) was instrumental in quantifying the immense energy cost and associated carbon footprint of these models, popularizing the crucial distinction between "Red AI" (computationally expensive, record-breaking models) and "Green AI" (research focused on high performance at a lower computational and environmental cost). The scaling laws that govern transformers, which have so successfully driven progress, also dictate that performance improvements often require an exponential increase in parameters and training data. This trajectory is fundamentally unsustainable. It not only limits the deployment of powerful models to a few large corporations but also creates significant barriers to equitable access.

Neuromorphic computing, and Spiking Neural Networks (SNNs) in particular, have emerged as a leading paradigm for Green AI, offering a path away from the brute-force computational paradigm of conventional systems. As detailed by pioneers in the field and summarized by Roy et al. (2019), SNNs derive their profound efficiency from two core, brain-inspired principles. First is event-driven computation: traditional Artificial Neural Networks (ANNs) operate on a dense, clock-based cycle, performing billions of multiply-accumulate (MAC) operations at every time step, regardless of whether the input has changed. This process is fundamentally inefficient. SNNs, in contrast, are event-driven. They compute only when a "spike"—a discrete event in time—arrives at a neuron, meaning processing is sparse and data-driven; no energy is expended on parts of the network that are not currently handling new information. This principle is analogous to the brain's own efficiency, where only a fraction of neurons are active at any given moment.

Second is temporal and structural sparsity: in a biological brain, and in a well-optimized SNN, neural activity is sparse. This sparsity is the primary driver of energy efficiency. A single synaptic operation (SOP) in an SNN, representing the processing of a single spike, is orders of magnitude less energy-intensive than the dense MAC operation that forms the basis of ANNs. This efficiency allows neuromorphic hardware, such as the Intel Loihi chip (Davies et al., 2021), to perform complex tasks at a fraction of the power consumption of a traditional GPU, which in turn enables secure on-device cognition and helps shrink the digital divide.

The STAC framework attempts to harness this revolutionary efficiency. It explores two distinct and complementary pathways to translate the proven computational power of transformers into the energy-efficient paradigm of SNNs. By doing so, it provides the essential missing piece of the puzzle: a computationally viable foundation upon which a truly personalized, multi-agent cognitive architecture like SCANUE can be built and deployed at scale. Without such a foundation, the goal of personalized AI for all remains an expensive and inaccessible dream.

Neuro-Psychometric Foundations of Personalized Alignment

To address the alignment challenge, the presented framework operationalizes critical insights from cognitive neuroscience and psychometric research. Specifically, the Synthetic Cognitive Augmentation Network Using Experts (SCANUE) architecture implements a neurobiologically plausible cognitive model inspired directly by the well-documented functional subdivisions within the human prefrontal cortex (Kennedy, Mohammad, & Wyandt, 2025; Tate, 2024a). Instead of a singular monolithic model, SCANUE strategically decomposes cognitive processing across five distinct, specialized, and computationally lightweight agents. Each agent emulates functions associated with specific PFC subregions extensively characterized in cognitive neuroscience literature: the dorsolateral prefrontal cortex (DLPFC) agent for executive control; the ventromedial prefrontal cortex (VMPFC) agent for emotional regulation and value-based decision-making; the orbitofrontal cortex (OFC) agent for reward processing; the anterior cingulate cortex (ACC) agent for conflict monitoring; and the medial prefrontal cortex (mPFC) agent for social cognition (Damasio, 1994; Miller & Cohen, 2001). To achieve this high degree of specialization, each agent was fine-tuned on domain-specific datasets relevant to its designated cognitive function. This modularization enhances interpretability and facilitates highly targeted user personalization.

This personalization is instantiated and maintained via the SCANAQ Alignment Questionnaire, a bespoke psychometric assessment composed of 36 carefully selected items. These items derive from eight validated psychological scales, each chosen for its relevance to the particular PFC functions modeled by the SCANUE agents (Tate, 2024c). The chosen scales encompass executive functions (BRIEF-A), emotional regulation (ERQ), impulsivity (BIS-11), risk-taking propensity (GRiPS), decision-making style (GDMS), self-efficacy (GSES), perceived stress (PSS), and empathy (IRI). In practice, SCANAQ serves as a translation layer, effectively converting a user's psychometric profile into actionable control signals that drive the dynamic behavior of SCANUE's specialized agents. This design ensures a theoretically grounded and empirically informed alignment between the user and the AI.

STAC: A Two-Phase Approach to Neuromorphic Efficiency

The energy-efficiency imperative, which is central to making a personalized system like SCANUE viable, is addressed through the Spiking Transformer Augmenting Cognition (STAC) framework. The development of STAC evolved through two distinct and methodologically significant phases, representing a deep exploration of two different pathways to neuromorphic efficiency.

The first phase, STAC V1 (Tate, 2025a), established a complete, end-to-end differentiable pipeline for a hybrid SNN-transformer, integrating a pre-trained GPT-2 backbone with custom spiking layers. This initial approach was highly innovative, moving beyond simple SNN layers to incorporate learnable AdEx (Adaptive-Exponential) spiking neurons and a custom Hyperdimensional Memory Module (HEMM). This design allowed the model to integrate biologically plausible dynamics and memory structures directly into the architecture. Energy efficiency was not an afterthought but was actively encouraged during fine-tuning through an L1 spike regularization term. The maturity of this version was confirmed by a comprehensive validation suite of seven distinct test functions. Its single-turn conversational capability was an intentional and logical consequence of its sequential, token-by-token processing architecture, not an unforeseen flaw.

While STAC V1 was a successful, self-contained research project, its immense computational cost and the inherent difficulty of scaling its bespoke training-based approach to compete with state-of-the-art LLMs prompted a strategic methodological shift. The research pivoted from a bespoke training-based approach to a more pragmatic, conversion-based one, motivated by the desire to leverage the power and accessibility of the vast ecosystem of existing pretrained models. This pivot aimed to make neuromorphic benefits applicable to them directly.

STAC V2 (Tate, 2025b) was therefore re-architected as an experimental conversion framework with the primary goal of translating conventional, pretrained transformer models into functionally equivalent SNNs. This approach trades the deep architectural integration and learnable dynamics of V1 for the practical advantage of wider applicability. The key innovation in V2 is the novel Temporal Spike Processor (TSP), a state-management module designed to preserve conversational context across multiple turns—the very capability not prioritized in V1's architectural paradigm. It is critical to note, however, that STAC V2 remains an experimental research prototype with known limitations and incomplete hardware operator coverage; its performance metrics are currently based on theoretical projections that await rigorous empirical hardware validation.

Contributions and Structure of the Paper

This manuscript synthesizes psychometric methodologies, cognitive neuroscience principles, and neuromorphic engineering into a cohesive framework for personalized cognitive augmentation. It frames the integration of the SCANUE cognitive architecture with the STAC neuromorphic backend as a primary future objective, thereby defining a clear roadmap from the current conceptual framework to a fully implemented system. Three innovations encapsulate the primary contributions: (1) a validated psychometric-to-agent mapping methodology that defines how to systematically translate SCANAQ-derived user profiles into dynamic agent control signals; (2) a detailed comparative analysis of two distinct neuromorphic strategies—STAC V1's hybrid fine-tuning approach and STAC V2's pragmatic conversion framework—highlighting the introduction of the Temporal Spike Processor (TSP) for multi-turn coherence in the latter; and (3) the establishment of a detailed, preregistered protocol for future validation that outlines rigorous benchmarks for alignment, cognitive efficacy, and energy efficiency assessments using specialized neuromorphic hardware.

Subsequent sections expand systematically on the framework's neuroscientific underpinnings, its methodological foundations, the detailed STAC development trajectories, the proposed validation protocols, and a critical discussion of current limitations, ethical considerations, and structured directions for future research.

BACKGROUND

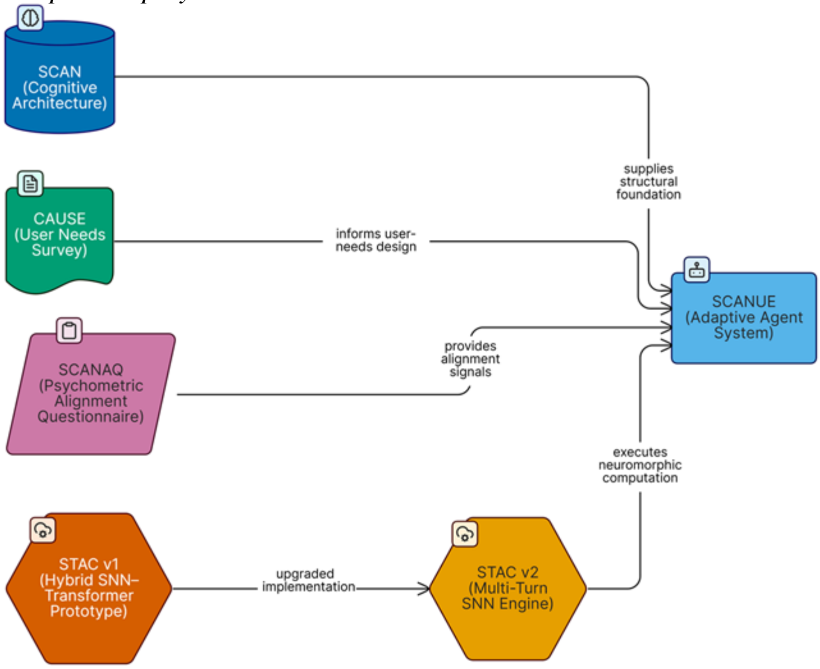
SCAN, SCANUE, and STAC in Context: A Multi-Layered Architecture

The conceptual evolution that produced the current integrated framework progressed through several key stages, beginning with the Synthetic Cognitive Augmentation Network (SCAN) (Tate & Sanders, 2024). SCAN introduced a novel, modular cognitive architecture designed to emulate the functional organization of the human prefrontal cortex (PFC). Marking a significant departure from monolithic LLM architectures, SCAN's foundational structure comprised five distinct expert agents corresponding to discrete PFC subregions (DLPFC, VMPFC, OFC, ACC, and mPFC), each utilizing a pretrained GPT-style backbone.

Building upon this foundation, the Synthetic Cognitive Augmentation Network Using Experts (SCANUE) framework (Tate, 2024a) introduced the critical enhancement of agent specialization through fine-tuning. In this version, each agent's pre-trained backbone was fine-tuned on domain-specific datasets, sharpening its function from a generalized expert into a specialized cognitive processor. The architecture was further refined in a subsequent iteration, SCANUE V22 (Tate, 2024b), which incorporated dynamic Human-in-the-Loop (HITL) feedback mechanisms. This version implemented memory-based loops for ongoing personalization, enabling the system to iteratively adjust agent outputs and progressively align its behavior with evolving user needs. This progression transformed the static computational model into an adaptive system capable of enhancing personal relevance and user trust.

Complementing this cognitive architecture, the Spiking Transformer Augmenting Cognition (STAC) framework supplies the essential computationally efficient execution layer. This research progressed through two distinct phases: STAC V1 (Tate, 2025a), a hybrid model that fine-tunes a pre-trained transformer and SNN model, and STAC V2 (Tate, 2025b), an experimental conversion prototype. The current STAC V2 framework is an experimental research prototype, limited to software-level SNN simulations, with hardware deployment and empirical energy measurements identified as key areas for future work. Additionally, SCANUE's broader architectural refinements benefited from the Cognitive Augmentation User Survey & Evaluation (CAUSE) (Tate, 2024c), a comprehensive internal psychometric tool designed to capture nuanced user needs and expectations for cognitive augmentation applications.

Figure 1. Integrated Framework Architecture: From Psychometric Assessment to Neuromorphic Deployment



Note. The diagram provides a schematic overview of the proposed integrated architecture. The process begins with user profiling via the SCANAQ instrument, which feeds into the adaptive orchestration of SCANUE and culminates in planned deployment on neuromorphic hardware via the STAC V2 pipeline. Solid lines denote active development trajectories, while dashed lines represent superseded legacy components.

Psychometric Foundation: The SCANAQ Alignment Questionnaire

Personalized alignment within the SCANUE framework originates from the administration of the Synthetic Cognitive Augmentation Network Alignment Questionnaire (SCANAQ) (Tate, 2024c). This specialized psychometric battery consists of a rigorously curated subset of 36 strategically selected items derived from eight established, psychometrically validated psychological instruments. These instruments were chosen for their theoretical relevance to distinct cognitive and emotional functions modeled by the five SCANUE agents. To ensure methodological transparency and reproducibility, the complete SCANAQ instrument is provided in Appendix A, and the comprehensive scoring breakdown is detailed in Appendix B.

SCANAQ's composite structure intentionally balances psychometric rigor with practical considerations for user engagement and assessment feasibility. Administering the entirety of all eight source psychological assessments would be impractical and burdensome, likely reducing user adoption. By meticulously selecting items with optimal predictive validity and reliability, SCANAQ preserves essential psychometric robustness while significantly enhancing participant feasibility, both during initial system onboarding and during periodic recalibration. Moreover, SCANAQ's development adhered to established cross-cultural adaptation guidelines (Beaton et al., 2000; Gudmundsson & Lönner, 2009). Such meticulous adherence ensures conceptual and psychometric equivalence across diverse linguistic and cultural populations, a fundamental requirement for any equitable global implementation strategy.

Neuroscientific Rationale for Agent Specialization and SCANAQ Integration

The rationale for the functional specialization across the five SCANUE agents and their direct integration with SCANAQ-derived psychometric metrics is grounded firmly in decades of established cognitive neuroscience research. The framework avoids ad-hoc design by creating systematic, theoretically informed mappings between validated psychological constructs and the behaviors of agents that emulate distinct functional regions of the prefrontal cortex (PFC). This grounding ensures that the personalization is not superficial but is instead rooted in an empirically robust understanding of human cognition. The following sections detail the neuroscientific and psychometric justification for each of the five agents.

DLPFC Agent: Executive Control and Goal-Directed Behavior

Neuroscientific Grounding. The dorsolateral prefrontal cortex (DLPFC) is widely recognized as the primary neuroanatomical region responsible for executive functions like planning and cognitive control. These higher-order processes include abstract reasoning, strategic planning, the online maintenance and manipulation of information in working memory, and top-down cognitive control (Miller & Cohen, 2001). It is the brain's "executive," responsible for orchestrating thought and action in accordance with internal goals. Foundational work by Petrides (2005) and others has detailed its critical role in tasks that require monitoring and manipulating multiple pieces of information to guide complex, multi-step behaviors.

Psychometric Justification. To precisely modulate this agent's behavior, the SCANAQ incorporates items from two key instruments: the Behavior Rating Inventory of Executive Function (BRIEF-A; Gioia et al., 2000) and the General Self-Efficacy Scale (GSES; Schwarzer & Jerusalem, 1995). The BRIEF-A is a gold-standard clinical and research instrument designed to assess the real-world behavioral manifestations of executive functions. Its subscales (e.g., Plan/Organize, Inhibit, Shift) map directly onto the cognitive processes governed by the DLPFC. The GSES, in turn, measures a user's domain-general belief in their ability to execute tasks and achieve goals, a core component of self-regulation tightly integrated with the persistence and effort components of executive control (Barbey et al., 2012). This pairing creates a direct, empirically validated, and clinically relevant link between a user's measured executive capabilities and the corresponding agent's supportive behavior.

To illustrate this principle, consider a user whose SCANAQ profile reveals a high T-score on the BRIEF-A's “Plan/Organize” subscale, indicating significant, self-reported difficulty in planning. If this user states a broad goal like, “I need to prepare for my annual performance review,” the DLPFC agent would provide an initial executive control response by presenting a structured delegation framework: *Subtasks*: 1. Compile major accomplishments from the past year. 2. Identify specific KPI examples. 3. Draft self-assessment narrative. *Agent Assignments*: VMPFC Agent will assess emotional readiness, OFC Agent will evaluate reward structures, ACC Agent will identify potential conflicts, mPFC Agent will ensure value alignment. *Integration Plan*: Each specialized agent would provide sequential analysis of their respective domains. The user would then receive additional sequential responses from each agent as it processes its assigned aspect of the task. While this transparent, sequential approach differs from the more implicit coordination of the biological DLPFC, it provides a computationally tractable approximation that delivers the cognitive scaffolding and systematic task decomposition the user's profile suggests they need.

VMPFC Agent: Emotional Regulation and Value-Based Decision-Making

Neuroscientific Grounding. The ventromedial prefrontal cortex (VMPFC) is a critical hub for integrating emotion into decision-making, regulating emotional responses, and assigning subjective value to potential choices (Damasio, 1994). Lesion studies, such as those involving the Iowa Gambling Task (Bechara et al., 2005), have famously demonstrated that damage to this area impairs the ability to use emotional signals to guide advantageous decision-making, even when knowledge of the risks is intact. The VMPFC allows us to have a “gut feeling” about what is good or bad for us, a process essential for navigating complex social and personal landscapes (Li et al., 2021).

Psychometric Justification. The VMPFC agent's behavior is primarily informed by three SCANAQ scales: the Emotion Regulation Questionnaire (ERQ; Gross & John, 2003), the General Decision-Making Style (GDMS; Scott & Bruce, 1995), and the Perceived Stress Scale (PSS; Cohen et al., 1983). The ERQ directly measures a user's habitual tendency to use two distinct emotion regulation strategies:

cognitive reappraisal (reframing a situation to change its emotional impact) and expressive suppression (inhibiting outward signs of emotion). These strategies are core VMPFC-mediated functions. The GDMS provides insight into the user's reliance on intuitive versus rational decision styles, a process heavily mediated by the VMPFC's valuation signals. Finally, the PSS quantifies the user's current cognitive and affective load, which is known to heavily impact VMPFC-mediated decision-making and emotional control.

Imagine a user with a SCANAQ profile indicating a high score on the ERQ's "Suppression" subscale and high concurrent perceived stress (PSS). This pattern suggests the user tends to bottle up negative emotions, a strategy that is cognitively costly and often maladaptive. If this user expresses significant anxiety about an upcoming public speaking engagement, the VMPFC agent would activate proactive affective scaffolding. It would avoid simplistic platitudes ("Don't worry, you'll be great!"). Instead, it would gently guide the user through a cognitive reappraisal exercise, asking reflective questions like, "I hear that you're feeling anxious. What is an alternative way to view this challenge not as a threat, but as an opportunity to share your expertise?" or "What aspects of this situation are fully within your control right now? Let's focus our attention on those." This intervention directly supports a healthier, more effective regulation strategy and pinpoints controllable factors the speaker can influence.

OFC Agent: Reward Processing and Outcome Evaluation

Neuroscientific Grounding. The orbitofrontal cortex (OFC), closely related to the VMPFC, is specialized for computing the value of expected outcomes and rapidly updating those values based on new information (Rolls, 2023). It is critical for flexible, adaptive behavior, allowing an organism to change choices when the reward contingencies of the environment shift. The OFC helps us learn from the consequences of our actions and adjust future behavior accordingly, playing a key role in reversal learning and in evaluating potential rewards versus risks (Schoenbaum et al., 2011).

Psychometric Justification. The OFC agent is calibrated by two SCANAQ scales: the General Risk Propensity Scale (GRiPS; Zhang et al., 2019) and the General Decision-Making Style (GDMS; Scott & Bruce, 1995). The GRiPS is a modern, validated measure of an individual's general tendency to take or avoid risks across various domains, directly tapping into the risk-benefit calculations central to OFC function. The GDMS complements this by revealing whether a user's decision style is more spontaneous and intuitive versus dependent and avoidant, which in turn informs how risk information should be framed.

For instance, for a user whose profile shows a very high score on the GRiPS, indicating a strong propensity for risk-taking, the OFC agent would adopt a detailed risk-communication strategy. If this user were considering a high-risk financial decision, such as investing a large sum in a volatile asset, the agent would not simply present the option. It would proactively annotate the choice with detailed probability information, meticulously delineate the potential best-case, worst-case, and most-likely scenarios, and transparently highlight the potential negative consequences. This approach would ensure the user's decision is fully informed, counterbalancing their natural propensity for risk with a clear-eyed view of the potential outcomes while preserving their autonomy.

ACC Agent: Conflict Monitoring and Error Detection

Neuroscientific Grounding. The anterior cingulate cortex (ACC) is often described as the brain's "conflict monitor." It becomes highly active during tasks that involve uncertainty, response conflict (e.g., the Stroop task), and the detection of errors (Botvinick et al., 2004). The ACC does not necessarily resolve the conflict itself but signals its presence to other brain regions, most notably the DLPFC, thereby recruiting more cognitive control to overcome the challenge. It is crucial for self-regulation, allowing us to catch and correct our mistakes and to proceed with caution in ambiguous situations (Kim et al., 2011).

Psychometric Justification. The ACC agent's behavior is dynamically informed by the Barratt Impulsiveness Scale (BIS-11; Patton et al., 1995). The BIS-11 is a widely used measure of impulsivity, a personality trait characterized by a tendency to act on a whim with little forethought. High impulsivity is strongly correlated with deficits in error monitoring and response inhibition—the very functions mediated by the ACC. Therefore, a user's BIS-11 score provides a direct, validated proxy for their likely need for enhanced conflict monitoring support.

In a hypothetical scenario, a user with a high score on the BIS-11's "Motor Impulsiveness" subscale would likely make quick, potentially irreversible decisions. If this user were to attempt to execute such an action within the system (e.g., "Delete all files in the archive folder"), the ACC agent would trigger a reflective-pause protocol. This protocol would extend far beyond a simple "Are you sure? [Y/N]" prompt. Instead, it would require a rationale from the user. The agent might respond, "This action will permanently delete 472 files and cannot be undone. To proceed, please briefly state your reason for deleting these files." This simple "speed bump" forces a moment of reflection to override a potentially impulsive and regrettable action, mirroring a neuro-cognitive conflict resolution process.

mPFC Agent: Social Cognition and Empathy

Neuroscientific Grounding. The medial prefrontal cortex (mPFC) is a cornerstone of the "social brain." It is essential for social cognition, which encompasses abilities including thinking about the mental states of others (Theory of Mind), understanding social norms, and self-reflection (Amodio & Frith, 2006). The mPFC allows us to step outside our own perspective to infer the beliefs, desires, and intentions of those around us, a skill fundamental to successful social interaction and empathy (Sullivan et al., 2022).

Psychometric Justification. The mPFC agent's functionality is modulated by the Interpersonal Reactivity Index (IRI; Davis, 1983). The IRI is a classic, multi-dimensional measure of empathy, assessing both cognitive components (like "Perspective-Taking," the ability to imagine others' viewpoints) and affective components (like "Empathic Concern," the tendency to experience feelings of sympathy). A user's profile on the IRI provides a nuanced picture of their empathic strengths and weaknesses, allowing for highly targeted social-cognitive support.

Consider a user with a low score on the IRI's "Perspective-Taking" subscale who is discussing an interpersonal conflict at work. This user might state, "I can't believe my colleague said that in the meeting. It was completely out of line." The mPFC agent, recognizing the user's difficulty in seeing the situation from another's point of view, would integrate reflective, socially contextual questions into the dialogue. Instead of reinforcing the user's anger, it might gently prompt, "That sounds frustrating. To help me understand the full picture, can we explore for a moment how your colleague might have perceived the situation?" or "What do you think their intention might have been, even if the impact was negative?" This approach would gently facilitate improved social cognition and empathetic understanding, scaffolding a skill the user may find challenging while still acknowledging their feelings.

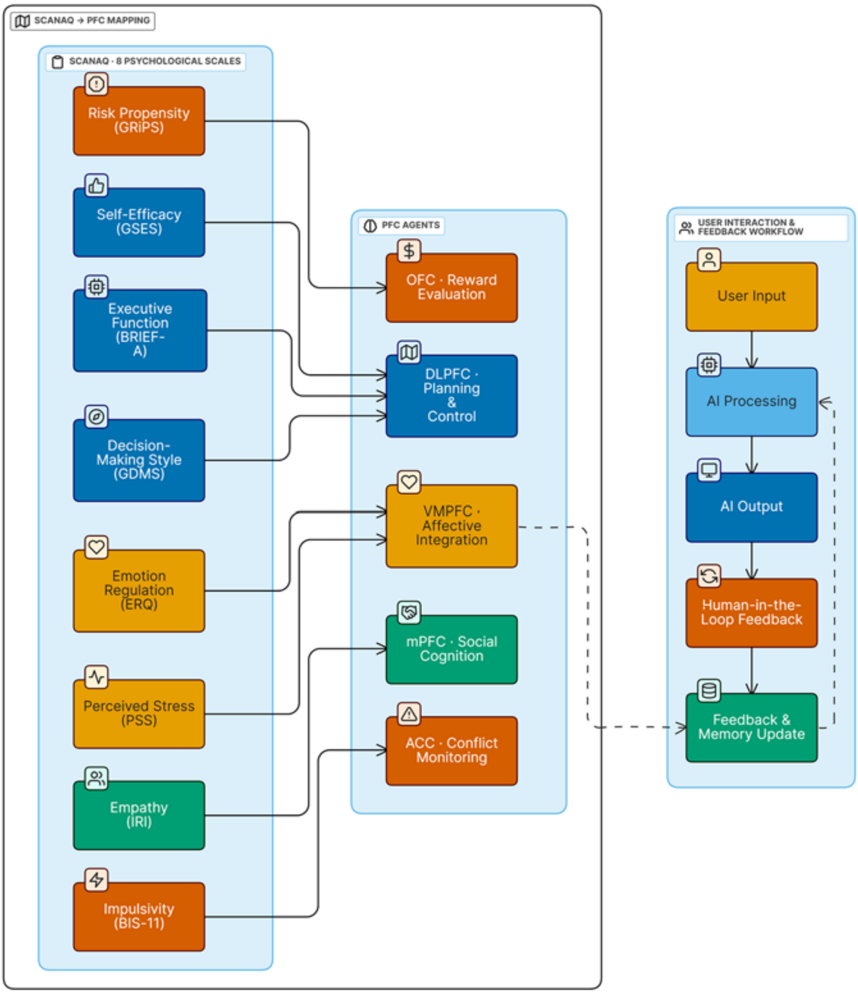
METHODOLOGICAL FOUNDATION: BRIDGING PSYCHOMETRICS, NEUROSCIENCE, AND AI

Translating SCANAQ Scores into Agent-Specific Behavioral Controls

The personalization mechanism of SCANUE involves clearly defined steps: (1) administering SCANAQ, (2) interpreting psychometric scores, and (3) translating these scores into instructions that dynamically adjust each specialized agent's behaviors. The numerical scoring breakdown and its mapping to specific PFC agent suggestions are available in Appendix B. A pivotal methodological strength of this approach is its deliberate preservation of psychometric nuance. Instead of applying broad normalization methods such as converting scores into generalized z-scores—which can obscure essential, instrument-specific details—the framework maintains the original Likert-type scaling intrinsic to each constituent measure (Tate, 2024c). By preserving the original scaling integrity, SCANUE ensures the nuances intended by the original scale developers remain meaningful in agent-driven applications. Future harmonization will aim to unify ranges without sacrificing this diagnostic signal.

Each SCANUE agent is selectively programmed to interpret and respond specifically to psychometric scores from relevant subsets of SCANAQ, chosen for their theoretical and neuroscientific alignment with the agent's specialized cognitive functionality. These scores translate into agent behaviors through detailed instructional mappings, which are designed as adaptive heuristics that shape nuanced agent responses. The entire workflow is visually conceptualized in Figure 2.

Figure 2. Psychometric-to-Agent Mapping and Dynamic Alignment Workflow



Note. This figure demonstrates the proposed initial mapping of SCANAQ-derived psychometric profiles onto their corresponding SCANUE agents and illustrates the dynamic interaction loop that facilitates the iterative, adaptive refinement of personalized alignment.

Dynamic Alignment Feedback Loop

SCANUE's dynamic alignment process is designed to recognize and accommodate the inherent variability in user states and preferences over time. The proposed personalized alignment process initiates upon completion of the SCANAQ assessment, which generates precise baseline agent configurations derived directly from psychometric scores. This step establishes an initial alignment foundation, which is then refined through the framework's currently implemented human-in-the-loop feedback mechanisms. Although this command-line interface integration is not yet implemented in the codebase, the validated SCANAQ provides a clear and actionable pathway for this future work.

During real-time interactions, dynamic collaboration among the five SCANUE agents is facilitated by advanced orchestration mechanisms like LangGraph. LangGraph systematically manages complex conditional logic, inter-agent communication, and agent-selection processes to determine precise agent engagement at any moment. This design supports flexible information flow, enabling investigations into alternative agent-routing options and diverse methods of use that are not constrained to a single biological analogue. Presently, each agent leverages specialized GPT-based models fine-tuned on domain-specific datasets that align with its cognitive functions (Kennedy et al., 2025; Tate, 2024c). A critical future development goal is the transition of these agents onto STAC's energy-efficient SNN computational substrates to enhance sustainable deployment capabilities.

To detect misalignment continuously, the framework's comprehensive telemetry systematically records explicit user feedback, including user-generated ratings and correctional interactions. Future enhancements will augment this process with more subtle data streams. These will include sophisticated analyses of implicit behavioral indicators—such as response latencies, editing behaviors, and sentiment fluctuations—alongside optional biometric integrations (e.g., EEG, fNIRS, heart rate variability, galvanic skin responses). With ethically secured user consent, such integrated data could significantly improve real-time assessments of cognitive load and stress levels.

Beyond detection, future framework extensions will incorporate adaptive recalibration mechanisms to correct for sustained misalignment. This recalibration would employ efficient, computer-adaptive short-form SCANAQ variants, administered dynamically based on interaction history to maintain alignment without the burden of full reassessment. As validated extensively in psychometric literature (Ling et al., 2022), these adaptive methodologies can refine agent behavioral parameters precisely and responsively, thereby enhancing long-term alignment with minimal user burden.

PROPOSED VALIDATION STRATEGY

Alignment and User Experience Metrics

Future evaluations will quantify alignment accuracy using primary metrics: frequency of misalignment events (target $\leq 5\%$), standardized usability scores (SUS ≥ 85), and structured qualitative analysis of user feedback to confirm meaningful alignment. To strengthen these assessments, a mixed-method approach combining quantitative and qualitative data collection is recommended. Quantitative analyses will include statistical modeling to identify and predict factors associated with alignment discrepancies, enabling proactive adjustments. This mixed-methods design will couple statistical power with narrative insights.

Qualitative methodologies will involve structured user interviews and focus groups aimed at capturing deeper insights into user perceptions and experiences. For instance, thematic analysis of interview transcripts could reveal nuanced user attitudes toward agent interactions, identifying areas where agent behavior aligns exceptionally well or consistently misses expectations. This qualitative component will facilitate targeted refinement of SCANAQ-derived mappings and enhance the precision of personalization.

Furthermore, additional empirical validation could leverage longitudinal study designs to assess alignment stability over extended periods. Regular, brief reassessments using computer-adaptive testing forms of SCANAQ would monitor alignment trajectories, ensuring sustained personalized efficacy without excessive user burden. Analyzing alignment data longitudinally can help identify emerging trends or gradual drift in agent-user congruence, which would trigger timely recalibration. Finally, including diverse participant populations in validation studies will ensure broader generalizability and equitable applicability. Attention must be given to capturing varied demographic backgrounds, cognitive styles, and professional roles to reinforce the robustness and inclusivity of validation findings.

Task-Specific Cognitive Outcomes

Agent efficacy will be systematically assessed using validated neuropsychological benchmarks. Classic tasks such as the Tower of London task (assessing planning capabilities, targeting DLPFC functions) and the Iowa Gambling Task (assessing risk-related decision-making, targeting OFC/VMPFC functions) will provide precise cognitive performance benchmarks.

Affective Congruence and Regulation Support

Structured stress-induction protocols, such as cognitive tasks under controlled time constraints, will be used to evaluate affective regulation efficacy by comparing predicted versus actual emotional responses. These assessments will critically examine the effectiveness of VMPFC-delivered emotional scaffolding using objective physiological and subjective self-report measures, with a target of significantly attenuating stress markers.

Energy Efficiency and Computational Performance

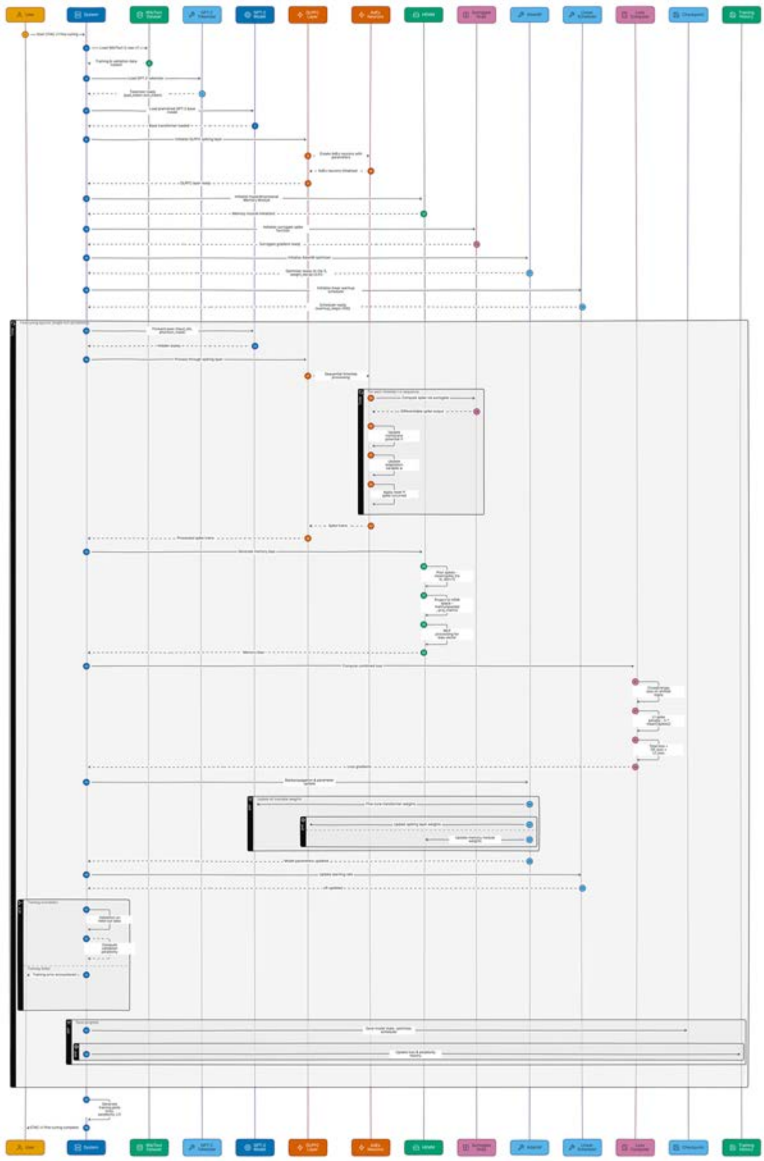
Rigorous benchmarking on specialized neuromorphic platforms, such as Intel's Loihi 2, will quantitatively validate computational efficiency claims. This step is vital for establishing credibility, as it moves beyond simulation to physical measurement. Precise spike-count telemetry and watt-hour energy consumption measurements will adhere to established neuromorphic benchmarks, transparently confirming (or refuting) theoretical projections empirically on Loihi 2 hardware.

STAC DEVELOPMENT TRAJECTORY: A METHODOLOGICAL EVOLUTION

STAC V1: A Deep Dive into the Hybrid Fine-Tuned Architecture

The initial version, STAC V1 (Tate, 2025a), was a landmark project designed to prove the fundamental viability of creating a high-performance hybrid SNN-transformer. Its architecture, illustrated in Figure 3, was a meticulous exercise in neuro-inspired engineering, defined by three key technical innovations observable in the project's implementation artifacts.

Figure 3. STAC V1 Hybrid Fine-Tuning Pipeline: Integrating a Pre-Trained Base with Spiking Component



Learnable AdEx Neurons and Surrogate Gradients.

The choice of the Adaptive-Exponential (AdEx) integrate-and-fire neuron model, based on the foundational work of Brette & Gerstner (2005), was deliberate.

Unlike simpler Leaky Integrate-and-Fire (LIF) neurons, the AdEx model captures a richer repertoire of biologically plausible dynamics, including spike-frequency adaptation and bursting. Its dynamics are governed by several parameters, such as the membrane time constant (τm) and the adaptation time constant (τw). The crucial innovation in STAC V1 was making these parameters learnable by defining them as torch.nn.Parameter tensors. However, the all-or-none spiking mechanism is non-differentiable, which poses a major obstacle to gradient-based training. STAC V1 solved this problem by using the surrogate gradient technique (Shrestha & Orchard, 2018; Zenke & Vogels, 2021). During the backward pass, this technique replaces the true, non-differentiable gradient of the spike-generation function with a "surrogate"—a continuous, well-behaved proxy like a fast sigmoid function. This proxy provides a useful learning signal that allows gradients to flow back through the spiking neurons, enabling the backpropagation algorithm to fine-tune not just the synaptic weights but also the intrinsic firing properties of the neurons themselves.

The Hyperdimensional Memory Module (HEMM).

To augment the model's ability to handle dependencies beyond immediate firing dynamics, STAC V1 introduced the HEMM, a structured memory system that departs from standard attention mechanisms. Its implementation follows a clear, efficient sequence. First, the full spike train tensor from the AdEx neuron layer is pooled across the time dimension via a torch.mean operation, creating a static vector that represents the sequence's overall spiking activity. This vector is then projected into a very high-dimensional space via matrix multiplication with a fixed, random projection matrix. This technique, inspired by hyperdimensional computing, creates a sparse, distributed representation of the activity history. This high-dimensional vector is then passed through a small multi-layer perceptron (MLP) to generate a final "memory bias," which is added back to the token representations before they enter the spiking layer. This mechanism allows the model's own recent past activity to influence its current processing in a structured, recurrent, and computationally efficient manner.

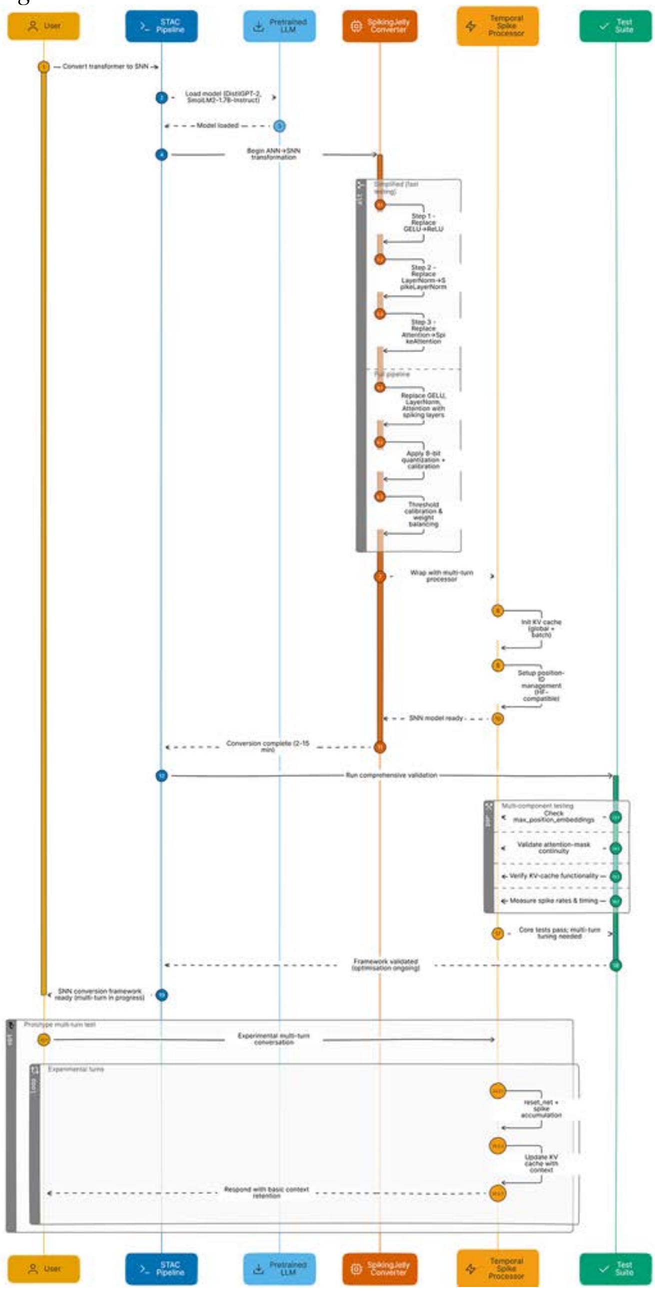
Integrated L1 Spike Regularization.

A core design principle of STAC V1 was that energy efficiency should be an integral part of the learning process, not a post-hoc analysis. To achieve this goal, the model actively encouraged sparse spiking activity by incorporating an L1-loss term directly into the main fine-tuning objective function. The total loss was a weighted sum of the standard cross-entropy loss and a sparsity loss, defined as L_sparsity = λ * torch.mean(torch.abs(spike_tensor)), where λ is a weighting coefficient that controls the strength of the sparsity penalty. By penalizing the mean absolute value of the spike tensor, the optimizer is incentivized to find solutions where a maximal number of elements in the tensor are zero (i.e., no spike occurred). This directly promotes the temporal sparsity that is the hallmark of efficient SNNs, making energy efficiency an objective of the optimization itself.

STAC V2: A Pragmatic Pivot and the Coherence-Fidelity Trade-off

While STAC V1 demonstrated feasibility, its scaling limitations led to the pragmatic development of STAC V2, which aims to convert existing models for broader usability. The AI field is dominated by massive LLMs trained by large corporations on planetary-scale datasets. The hybrid fine-tuning paradigm, while academically elegant, could not realistically scale to these models. Recognizing this limitation, the research pivoted from deep integration to pragmatic conversion. The central motivation for STAC V2 (Tate, 2025b) was to develop a toolchain that could apply the energy-saving benefits of neuromorphic computing to the vast ecosystem of existing pretrained models, thereby democratizing access to neuromorphic efficiency. This toolchain is outlined in Figure 4.

Figure 4. STAC V2 Conversion Framework: From Pretrained Transformers to Stateful Spiking Networks



This pivot, however, introduces a well-known challenge in neuromorphic engineering: a trade-off between energy efficiency and computational fidelity. The degradation in conversational quality observed in STAC V2 is a direct and predictable consequence of the information loss inherent in the ANN-to-SNN conversion process.

Activations in a traditional ANN are represented by high-precision, continuous floating-point numbers. The STAC V2 conversion pipeline, relying on libraries like SpikingJelly (Fang et al., 2021) and principles from works such as Diehl et al. (2015), must perform a form of quantization. It must map this rich, continuous range of activation values onto a stream of discrete, binary spike events over time—an inherently lossy process. A small degree of error is introduced at every layer as continuous values are approximated by spike rates or timings. While the error at any single layer may be negligible, it compounds as data flows through the dozens of layers in a deep transformer architecture. This cumulative precision loss ultimately manifests as a noticeable degradation in the coherence and nuance of the final text output when compared to the original, full-precision ANN.

A critical area of future research, therefore, is the development of "quantization-aware" conversion techniques or post-conversion tuning strategies to mitigate this fidelity loss. Such methods might involve fine-tuning the converted SNN for a few epochs on a small dataset, allowing it to recover from the conversion noise, or employing more sophisticated conversion algorithms that can better preserve the information content of the original model. Acknowledging this trade-off is crucial for setting realistic expectations about the current state of SNN technology and for guiding future research toward building SNNs that are not only efficient but also powerful.

Preliminary Findings and Future Benchmarking

The development of STAC V2 was sharply focused on enabling multi-turn conversations through post-hoc conversion, a more pragmatic goal than V1's hybrid fine-tuning. Initial functional "smoke tests" were conducted to verify the new architecture's core hypothesis. These tests confirmed that V2's Temporal Spike Processor (TSP) (Tate, 2025b) can mechanically pass conversational memory states across successive interactions within a software simulation. This result demonstrates the feasibility of the basic architectural approach for enabling stateful interactions, though it is not a guarantee of coherent conversation.

It is essential to contextualize STAC V2's results with two critical caveats. First, its projected three- to four-fold energy savings are purely theoretical, derived from post-hoc spike telemetry; this contrasts with the integrated L1 regularization used to enforce sparsity during STAC V1's fine-tuning. Second, the conversion process embodies a significant trade-off: the resulting SNN, while stateful, suffers from a clear degradation in conversational coherence and nuance compared to its source

ANN. Blind A/B human-rating studies will be required to formally score coherence, helpfulness, and safety.

DISCUSSION

Integration of Psychometric Alignment and Neuromorphic Efficiency: A Synergistic Approach

This research presents a strategically integrated approach aimed at resolving two typically separate yet deeply interconnected challenges in contemporary AI: achieving deep personalized cognitive augmentation while simultaneously realizing significant computational and environmental efficiency through neuromorphic computing. The deliberate integration of psychometrically derived user profiles—captured through the validated SCANAQ assessment—with the specialized, prefrontal-cortex-inspired cognitive agents of the SCANUE architecture directly addresses ongoing challenges related to user trust, personal relevance, and the ethical deployment of AI systems (Kennedy et al., 2025). By linking validated psychometric metrics to agent-specific cognitive behaviors, the SCANUE-SCANAQ alignment strategy fosters precisely tailored interactions. This alignment ensures the assistant's communications and support strategies are continuously aligned with the nuanced cognitive and affective preferences unique to individual users, which in turn sustains engagement, builds trust, and ensures genuine utility in practical cognitive-augmentation contexts.

Simultaneously, the STAC V2 framework addresses computational sustainability by leveraging biologically inspired, event-driven SNNs. Crucially, STAC V2's innovations—particularly the introduction of the Temporal Spike Processor (TSP)—provide the mechanical foundation for stateful multi-turn interactions in software simulations. This architectural synergy supports the computational feasibility of deploying personalized cognitive agents and ensures practical scalability. Thus, SCANUE's advanced cognitive architecture provides meaningful, complex cognitive missions ideally suited to leverage STAC's computationally efficient execution capabilities. Conversely, STAC's neuromorphic efficiency makes the practical implementation of such sophisticated personalization computationally feasible at scale and justifies on-device deployment.

Limitations and Future Directions for Research and Refinement

While this work presents a promising synthesis, several limitations define the immediate trajectory for future research. On a technical level, the projected energy savings require empirical validation on dedicated neuromorphic hardware, as current operator coverage is limited and the conversion process can degrade performance on complex reasoning tasks. Scalability to larger transformer models also remains a significant hurdle. Concurrently, from a user-centric standpoint, critical questions persist regarding the stability of personalized alignment over long-term interaction and the framework's ability to generalize effectively across diverse populations. To launch this next phase of research, two initiatives are proposed: a one-month study to benchmark energy consumption on neuromorphic hardware and a three-month longitudinal trial to rigorously evaluate the durability of user alignment.

Ethical, Societal, and Governance Considerations: Navigating Responsible Innovation

Meticulous attention to ethical concerns, such as cognitive over-reliance and potential biases in psychometric assessments, is critical. Privacy and confidentiality must be proactively safeguarded, particularly because the SCANUE-SCANAQ framework relies on sensitive psychometric data. To prevent unauthorized data usage, robust data encryption, ISO-27001-grade storage protocols, and strict access control mechanisms should be implemented. Additionally, user autonomy and informed consent must be prioritized through transparent communication about how their data is collected, stored, and utilized.

The risk of cognitive over-reliance poses an ethical and practical challenge. While personalized cognitive augmentation systems aim to support user capabilities, users might become excessively dependent on AI-generated advice, diminishing their own decision-making confidence over time. To mitigate this risk, specific design features should be incorporated, such as periodic prompts encouraging independent critical reflection and clear communication of uncertainty in AI-generated recommendations. Continuous monitoring of user interactions could further help identify patterns of dependency, allowing for proactive interventions aimed at empowering users rather than diminishing their cognitive autonomy.

Potential biases inherent in psychometric assessments present another significant ethical consideration. Given SCANAQ's reliance on established psychological scales, it is essential to recognize and address biases that could affect fairness across diverse populations, including cultural biases and language barriers. Rigorous validation across diverse demographic and cultural groups is crucial. Standard practice should include regularly updating assessment items based on empirical feedback, incorporating culturally sensitive adaptations, and employing advanced statistical methods for bias detection (e.g., Differential Item Functioning analyses) to ensure fairness and inclusivity.

Moreover, governance considerations must include clear accountability frameworks. Given the complex interplay between AI, psychometric evaluation, and neuromorphic computing, a clear delineation of responsibilities and oversight mechanisms is necessary. Independent ethics review boards, comprising interdisciplinary experts, could regularly audit framework deployments, particularly in high-stakes contexts like healthcare or education. Transparent documentation, reporting standards akin to clinical trial registries, and public fail-case reports should be adopted to ensure public accountability and foster trust among stakeholders, regulators, and the broader community.

CONCLUSION AND FUTURE OUTLOOK

The integration of the SCANUE, SCANAQ, and STAC frameworks provides a viable architecture for developing sustainable, deeply personalized cognitive support systems capable of enhancing user trust, engagement, and equitable access to advanced AI. This paper has chronicled a principled research evolution—from the hybrid fine-tuning of STAC V1 to the more pragmatic, experimental conversion approach of STAC V2. By synthesizing principles from cognitive neuroscience, rigorous psychometric methodologies, and an iterative exploration of neuromorphic paradigms, this work offers a comprehensive roadmap for deploying the next generation of cognitive augmentation technologies.

While this synthesis is promising, the path to implementation requires addressing clear next steps. Technically, the framework's performance and efficiency projections must be empirically validated on specialized neuromorphic hardware and scaled to larger models. Concurrently, the long-term stability and generalizability of its user-centric alignment demand rigorous longitudinal evaluation across diverse populations.

Successfully navigating these research pathways will position the integrated framework for real-world implementation across critical domains, including clinical decision-making support, personalized adaptive education, and private, responsive edge-computing environments. Ultimately, achieving this integrated vision advances cognitive augmentation technologies that honor human individuality while responsibly addressing global sustainability imperatives.

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APPENDIX A

Synthetic Cognitive Augmentation Network Alignment Questionnaire (SCANAQ)

Introduction

Welcome to the *Synthetic Cognitive Augmentation Network Alignment Questionnaire* (SCANAQ). This assessment evaluates aspects of your cognitive functioning and personal preferences to align the SCAN system with your needs.

Instructions

1. Read each statement carefully.
2. Use the section's rating scale table to determine your response number.
3. Enter only the number in the Response column.
4. There are no right or wrong answers—respond based on typical thoughts, feelings, and behaviors.

Section A: Executive Functioning (Items 1–8)

Table 1.

Item	Statement	Response (1–3)
1	I have trouble getting started on tasks.	
2	I forget instructions easily.	
3	I have trouble keeping my emotions under control.	
4	I make careless mistakes.	
5	I get stuck on one way of doing things.	
6	I have trouble planning ahead.	
7	I have difficulty organizing my tasks.	
8	I have trouble keeping track of my belongings.	
Rating Scale 1- Never; 2- Sometimes; 3- Often		

Section B: Emotion Regulation (Items 9–12)

Table 2.

Item	Statement	Response (1–7)
9	I control my emotions by changing the way I think about the situation I'm in.	
10	When I want to feel less negative emotion, I change the way I'm thinking about the situation.	
11	I keep my emotions to myself.	
12	When I am feeling negative emotions, I make sure not to express them.	
Rating Scale 1- Strongly Disagree; 2- Disagree; 3- Somewhat Disagree; 4- Neither Agree nor Disagree; 5- Somewhat Agree; 6- Agree; 7- Strongly Agree		

Section C: Impulsivity (Items 13–18)

Table 3.

Item	Statement	Response (1–4)
13	I don't pay attention.	
14	I make up my mind quickly.	
15	I act on the spur of the moment.	
16	I buy things on impulse.	
17	I don't plan tasks carefully.	
18	I am more interested in the present than the future.	
Rating Scale 1- Rarely/Never; 2- Occasionally; 3- Often; 4- Almost Always/Always		

Section D: Risk Propensity (Items 19–21)

Table 4.

Item	Statement	Response (1–5)
19	I enjoy taking risks in general.	
20	I often take risks in my daily life.	
21	Taking risks is an important part of my life.	
Rating Scale 1- Strongly Disagree; 2- Disagree; 3- Neutral; 4- Agree; 5- Strongly Agree		

Section E: Decision-Making Style (Items 22–26)

Table 5.

Item	Statement	Response (1–5)
22	I make decisions in a logical and systematic way.	
23	I rely on my instincts when making decisions.	
24	I depend on others to help me make decisions.	
25	I avoid making important decisions until the pressure is on.	
26	I make quick decisions without thinking them through.	
Rating Scale 1- Strongly Disagree; 2- Disagree; 3- Neutral; 4- Agree; 5- Strongly Agree		

Section F: Self-Efficacy (Items 27–29)

Table 6.

Item	Statement	Response (1–4)
27	I can always manage to solve difficult problems if I try hard enough.	
28	If someone opposes me, I can find the means and ways to get what I want.	
29	I am confident that I could deal efficiently with unexpected events.	
Rating Scale 1- Not al all True; 2- Hardly True; 3- Moderately True; 4- Exactly True		

Section G: Perceived Stress (Items 30–32)

Table 7.

Item	Statement	Response (1–5)
30	In the last month, how often have you felt unable to control the important things in your life?	
31	In the last month, how often have you felt nervous and stressed?	
32	In the last month, how often have you felt that things were going your way?	
Rating Scale 1- Never; 2- Almost Never; 3- Sometimes; 4- Fairly Often; 5- Very Often		

Section H: Empathy & Social Cognition (Items 33–36)

Table 8.

Item	Statement	Response (1–5)
33	I often have tender, concerned feelings for people less fortunate than me.	
34	When I see someone being taken advantage of, I feel kind of protective towards them.	
35	I sometimes find it difficult to see things from the other person's point of view.	
36	I daydream and fantasize, with some regularity, about things that might happen to me.	
Rating Scale 1- Does Not Describe Me Well; 2- Describes Me a Little; 3- Somewhat Describes Me; 4- Describes Me Well; 5- Describes Me Very Well		

Completion Notes

Thank you for completing the SCANAQ. Your responses will help us generate a personalized alignment profile to enhance your interaction with SCAN.

- This is not a diagnostic tool.
- Participation is voluntary and confidential.
- All data is securely handled.

APPENDIX B

SCANAQ Scoring Breakdown, Descriptions, and PFC-Inspired Suggestions

Section A: Executive Function (EF) — Items 1–8

Table 9. Scoring / Interpretation

PROFILE CODE	PROFILE LABEL	SCORE RANGE	ITEMS (Q#)	DESCRIPTION	PFC-INSPIRED SUGGESTION
4A	High Inhibition, Low Planning	8–13	1–8	Difficulty initiating and planning tasks.	Provide planning tools, reminders, organizational aids; break tasks into smaller steps.
4B	Low Inhibition, High Working Memory	14–24	1–8	Strong working memory; impulsivity may need regulation.	Offer impulse-control strategies; leverage strong WM for complex/multistep tasks.

Section B: Emotion Regulation (ER) — Items 9–12

Table 10. Scoring / Interpretation

PROFILE CODE	PROFILE LABEL	SCORE RANGE	ITEMS (Q#)	DESCRIPTION	PFC-INSPIRED SUGGESTION
2B	Low Cognitive Reappraisal, High Expressive Suppression	4–19	9–12	Difficulty reframing emotions; relies on suppression strategies.	Train cognitive reappraisal via CBT-style reframing tools and guided cognitive restructuring.
2A	High Cognitive Reappraisal, Low Expressive Suppression	20–28	9–12	Strong emotional reframing; balanced expression.	Provide expressive-regulation tools to fine-tune social display and contextual appropriateness.

Section C: Impulsivity (IMP) — Items 13–18

Table 11. Scoring / Interpretation

PROFILE CODE	PROFILE LABEL	SCORE RANGE	ITEMS (Q#)	DESCRIPTION	PFC-INSPIRED SUGGESTION
5B	High Motor Impulsivity, Low Non-Planning Impulsivity	6–12	13–18	Acts impulsively; limited advance planning.	Embed pause-and-plan prompts; use external structure, checklists, and "if-then" implementation intentions.
5A	High Attentional Impulsivity, Low Motor Impulsivity	13–24	13–18	Distractible but less motorically impulsive.	Provide attentional control aids (mindfulness, focus timers, distraction blockers).

Section D: Risk Propensity (RP) — Items 19–21

Table 12. Scoring / Interpretation

PROFILE CODE	PROFILE LABEL	SCORE RANGE	ITEMS (Q#)	DESCRIPTION	PFC-INSPIRED SUGGESTION
6B	Low Risk-Taking	3–9	19–21	Avoids risk; prefers safer choices.	Scaffold calculated exploration; present graded exposure to uncertainty with feedback.
6A	High Risk-Taking	10–15	19–21	Seeks/enjoys taking risks.	Provide risk-assessment tools, consequence simulators, and decision hygiene checklists.

Section E: Decision-Making Style (DM) — Items 22–26

Determination Rule Table

Determine the **single highest scoring item** among Q22–Q26; assign the corresponding style below (no cumulative score).

Table 13. Style Interpretation

STYLE CODE	PRIMARY ITEM	STYLE LABEL	DESCRIPTION	PFC-INSPIRED SUGGESTION
1A	Q22	Rational	Logical, systematic decision-making.	Provide data-driven tools, structured analysis frameworks, and MCDA/ utility modeling.
1B	Q23	Intuitive	Relies on instincts.	Pair intuition with post-hoc analytical checks; use scenarios/ simulations for calibration.
1C	Q24	Dependent	Seeks guidance and collaboration.	Offer collaborative tools/decision aids; scaffold autonomy via staged independence.
1D	Q25	Avoidant	Procrastinates/avoids decisions.	Use proactive prompts, deadlines, and commitment devices; provide bounded-choice frameworks.
1E	Q26	Spontaneous	Quick, reflexive decisions with limited forethought.	Introduce pause rules ("10-second rule"), consequence visualizations, and pre-commitment plans.

Section F: Self-Efficacy (SE) — Items 27–29

Table 14. Scoring / Interpretation

Profile Code	Profile Label	Score Range	Items (Q#)	Description	PFC-Inspired Suggestion
7B	Low Self-Efficacy	3–8	27–29	Low confidence handling problems/ challenges.	Build mastery via graded goals, mentorship, success tracking, and strengths-based reflection.
7A	High Self-Efficacy	9–12	27–29	Confident and effective under challenge.	Leverage leadership potential; assign complex tasks; promote peer coaching/knowledge transfer.

Section G: Perceived Stress (PS) — Items 30–32

Table 15. Scoring / Interpretation

PROFILE CODE	PROFILE LABEL	SCORE RANGE	ITEMS (Q#)	DESCRIPTION	PFC-INSPIRED SUGGESTION
8B	Low Perceived Stress	3–9	30–32	Reports lower stress frequency.	Maintain resilience practices; add anticipatory coping plans for high-demand periods.
8A	High Perceived Stress	10–15	30–32	Frequently stressed/ overwhelmed.	Provide stress-management (CBT, relaxation, time-buffering, prioritization aids).

Section H: Empathy and Social Cognition (ESC) — Items 33–36

Table 16. Scoring / Interpretation

PROFILE CODE	PROFILE LABEL	SCORE RANGE	ITEMS (Q#)	DESCRIPTION	PFC-INSPIRED SUGGESTION
3B	Low Empathy	4–12	33–36	Difficulty relating to or understanding others.	Social skills training, perspective-taking exercises, and feedback on empathic accuracy.
3A	High Empathy	13–20	33–36	Strong empathic concern and social attunement.	Encourage mentoring/ leadership roles; integrate into cooperative, high-EQ team workflows.

Global Notes

Scores are summed within each section (except Decision-Making Style, which is determined by the single highest-scoring item among Q22–Q26).

These profiles are intended to **align and personalize** SCAN's agent behaviors; they are **not diagnostic categories**.

Ensure all processing complies with **privacy, ethical, and regulatory** standards (e.g., HIPAA, GDPR, IRB guidance as applicable).